

Question 1 - 0417/May/June/24/11 Q2

(a)	Encryption	1
(b)	Control	1
(c)	Hacking	1
(d)	(An) interview	1

Question 2 - 0417/May/June/24/11 Q10

	Four from: The microprocessor receives the data A preset value for the distance is stored in the microprocessor The microprocessor compares this data to the minimum distance/preset value If the data is less than the preset value the microprocessor sends a signal to the actuator The actuator applies the brakes If the data is more than the preset value the microprocessor does nothing Process is continuous	4
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Question 3 - 0417/May/June/24/11 Q8

(a)	Two from: Gaming Example of walk through systems Example of training systems	2
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(b)	Max Six from: People can be trained allowing them to practice over and over until they have perfected a particular task People can be trained in a virtual environment without risk to themselves/others Allows training in dangerous situations where it is impossible to practice the real thing Can give people a much more interactive experience when they are trained Allows people to retain knowledge much better than reading about it in a book/training room Allows people to walk through to check for potential errors/tested before the design is implemented Designs can be modified quickly therefore better objects can be made Confusion between real life and the game Causes headaches/eye strain/motion sickness/falling over Can become addicted to gaming with this system	6
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Question 4 - 0417/May/June/24/12 Q10

	Four from: Data are sent back to the sensor The sensors scan continually The sensor captures the reflected data The sensor captures analogue data The data from the sensor is sent by the ADC	4
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Question 5 - 0417/May/June/24/12 Q4

	Max five from: Positive Travel becomes safer Allows the driver to concentrate more on the driving Helps to prevent human error Keeps pedestrians and other people safe Can avoid accidents Max five from: Negative Become very reliant on the technology Very costly to repair/maintain Makes the vehicle more expensive The device/microprocessor could stop working/misread causing injuries Error messages/alerts could distract the driver	6
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Question 6 - 0417/May/June/24/12 Q8

(a)	Two from: Virtual Reality is completely computer generated world Virtual Reality may require a headset/goggles Virtual Reality only enhances a fictional reality Augmented Reality can be accessed with a smartphone Augmented Reality combines both the virtual and real world Augmented Reality creates a new layer on top of existing reality	2
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(b)	Six from: It can help people connect with others to improve communication skills Overlays information/warnings on the display therefore leads to safer activities Produces more efficient/engaging shopping as user could look at a product and information can be displayed The user can try on clothing electronically so you can see what the clothes look like (on you) before buying Used in walk through systems to give a greater experience for users Using it in (interactive) gaming could allow people to have a better experience of the game As it is used in gaming it could lead to safety issues as the player could be immersed in the game Could damage social relationships/skills as it removes face to face interaction On screen information could be distracting Allows us to interact with various characters/situations in real-time but not necessarily the real life	6
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Question 7 - 0417/Oct/Nov/23/12 Q11

(a)	Six from: <i>Benefits</i> • Saves the cost of having a store • Saves cost of rent/insurance/utilities/costs/business rates • Saves cost as less staff required • Larger customer base therefore increases profits • The selling is not limited to shop opening hours therefore increases sales <i>Drawbacks</i> • Increased costs more staff needed to operate/develop the website/drivers • Needs large warehouse space/distribution system • Reliant on the technology • Operates 24/7 therefore increases overheads • Relies on customers that only shop online To gain full marks the candidate must have both benefits and drawbacks	6
(b)	Five from: Customer: • Loads up the website • Sets up an account • Logs into the website • Enters a password • Browses through the online catalogue • Clicks on the items required • Enters the quantity/colours/size of items • Adds items to the virtual basket/cart • Clicks on the virtual checkout • Chooses payment method • Chooses delivery method • Adds payment/delivery/personal details • Checks the details entered • Logs out of system	5

Question 8 - 0417/May/June/23/12 Q13

(a)	Four from: The large number of variables makes it more difficult for a human to forecast correctly/accurately Can predict long range weather trends/global warming Faster way of producing weather patterns than using a human Can deal with the more complex calculations quicker More accurate results produced Can give residents early warning of severe weather	4
(b)	Two from: Pressure Temperature Moisture/Humidity Wind speed	2
(c)	Two from: Monitor Printer Plotter	2

Question 9 - 0417/May/June/23/12 Q14

(a)	Four from: - Allows multimedia to be embedded in the pages - Allows auto change of pages - It is digital so can be used in other documents - Sections can be linked internally/externally - Readers can interact with the ePublication - Can be used by many users simultaneously - More up-to-date than physical magazines	4
(b)	Two from: - Users type in a unique licence key/serial number/authorisation key - Users install DRM (parts of the code which stop you from copying it) - Use of licence agreements - Running the software with a dongle/CD/Pendrive attached - Getting permission from the owner	2

Question 10 - 0417/May/June/23/11 Q12

(a)	Two from: - Live or recorded content - Delivered to computers/mobile devices via the internet - Played back in real time/on demand. - Allows the user to view the media without downloading it	2
(b)	Four from: Benefits - Instant access to the media - Saves time rather than downloading/going to the shops - No need for storage space as there is no copy Drawbacks - Requires an internet connection - Requires a good internet/bandwidth connection otherwise buffering - Don't get to keep the product - Online security issue related to the subscription - Technology could fail in the middle of the stream - If media is downloaded it is time locked To gain full marks candidates must have correct answers for both benefits and drawbacks	4

Question 11 - 0417/May/June/23/11 Q8

(a)	Six from: Similarities Both read/scan data using optical methods Both use direct data entry Both are input devices Both read data from printed documents Both input large amounts of data Both read data very quickly Differences OMR does not need to read the characters written only the marks OMR is unable to recognise machine or hand-printed characters For OMR to read, marks need to be within a specified area The OMR marks have to be in the correct location OCR is more accurate at reading detailed data Data read from an OCR can be manipulated by word processing software OCR requires manual intervention to change the scanned words. To gain full marks candidates must have correct answers for both similarities and differences	6
(b)	Two from, for example: Lottery tickets Bar codes//QR codes Multiple Choice forms//Questionaries Attendance registers	2

Question 12 - 0417/May/June/23/13 Q3

(a)	Six from: Advantages Larger screen to read the typed data Laptop uses a physical keyboard therefore easier to type Larger keyboard to type in the data Fewer data entry errors due to larger screen / keyboard More suited for entering mass amounts of data Laptop version of software will have more features than smartphone version of software More computer storage for typed text Easier to add other devices to store the data Disadvantages Less portable / heavier than a smartphone as the smartphone can be slipped into a pocket Smartphones are smaller therefore easier to carry More likely to have a smartphone with them To gain full marks candidates must have correct answers from both advantages and disadvantages.	6
(b)(i)	Two from: Camera Microphone Touch screen	2
(b)(ii)	Two from: Speaker Screen	2
(b)(iii)	Two from: SD card Flash memory Internal storage SSD	2

(c)	Two from: Retains the format of the original newsletter Takes up less storage space Independent of hardware and software	2
(d)	Four from: Interesting / engaging content Use of longer sentences / not using slang / emoticons Use of some technical language More text than pictures Balanced use of white space and content Clear readable fonts Formal layout Balance of size between headings and text Contrasting colours Use suitable images More real images than cartoons	4

Question 13 - 0417/Oct/Nov/23/12 Q6

(a)	Two from: Eye problems Headaches Burns	2
(b)	Two from: Projector falling on the presenter Electrical fire due to overheating of the projector Electrocution caused by liquid spilling on the projector Tripping over trailing cables	2

Question 14 - 0417/May/June/23/13 Q7

(a)	Four from: By electronically tracking the sales of goods the industry could know when to order items... ...so that the goods do not run out in the stores The POS system allows a business to identify which products are selling well / selling badly... ...so products do not run out / left of the shelves Through this identification, a business can increase the order of profitable products and reduce unmarketable inventory Increases speed of transaction as the checkout staff do not need to enter the price of each item Itemised bills can be generated on screen and printed for customers It allows self check out which means less staff are needed therefore reducing staff costs	4
(b)	Six from: The (bar codes on) goods are scanned at the checkout The item code is searched in the stock database When the item is found... ...the number sold of that item is read The number in stock is reduced by this number If the number in stock is lower than the reorder / preset value... ...then a message is sent to automatically re-order stock When the new stock arrives database is updated with the new stock Quantity available will be increased when new stock arrives Generates a report on selling trends	6

Question 15 - 0417/May/June/23/12 Q9

(a)	Six from: Similarities Both readers scan the product code Both readers use direct data entry Both readers allow tracking of products Both readers are contactless Differences Bar code readers only scan one bar code at a time whereas RFID readers read multiple tags at the same time Bar code readers require a direct line-of-sight with the code RFID readers use near-field technology Bar code readers need to be close to the bar code Bar code needs to be in the correct position for reader to be able to read it Bar codes readers might not be able to read a damaged/scratched bar code whereas RFID can read data as long as chip/tag is not severely damaged. Bar code reader might have read errors if the bar code is damaged/scratched whereas RFID readers are more likely to read damaged/scratched tags Bar code readers emit/use light whereas RFID uses electromagnetic/radio waves To gain full marks the comparison must have correct answers for both similarities and differences.	6
(b)	Two from for example: Race timing Tracking Event Attendee Tracking Livestock Tracking Library book Tracking Contactless/credit/debit card payment Passports/ID cards	2